Even-odd Multigrid Solver: for Möbius Domainwall Fermion

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Note on the even-odd multigrid solver of RutticeQCD for Möbius domain-wall fermions.

This document derives what the MG note[1] ("the note") states. The note gives the method and its results in compact form; here each formula is derived with every symbol defined before it is used, so that the method can be followed with nothing more than standard graduate coursework in lattice QCD. The sections follow the note: each heading names the section of the note it derives (Secs. II–IV), so the two can be read side by side. The last section maps each step to the code. HMC, which the note leaves to future work, is derived in a separate walkthrough.[2] Nothing here is new with respect to the note and the code.

The solver in one page

The task is $D|\psi\rangle = |\eta\rangle$ for the Möbius operator at a light quark mass. The solver goes through the following steps; each later section derives one of them.

1. *Even-odd reduction* (Sec. 2.2). Eliminate the odd sites: $S|\psi_e\rangle = |\eta'_e\rangle$ on the even sites, then $|\psi_o\rangle = C_o^{-1}(|\eta_o\rangle - D_{oe}|\psi_e\rangle)$. This needs C^{-1} exact and cheap (Sec. 1.6).
2. *Pauli–Villars preconditioning* (Sec. 2.4). With $|\psi_e\rangle = C_e^{-1}|y\rangle$ and the factor $(S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger$ from the left, solve $\tilde{A}_R|y\rangle = |\rho_R\rangle$, whose spectrum lies in the right half-plane.
3. *Outer solver* (Sec. 7). Flexible GCR(2) on $\tilde{A}_R$ (or $\tilde{A}_L$) with one multigrid cycle $\mathcal{M}$ as right preconditioner per step.
4. *One cycle* (Sec. 6). A Chebyshev smoother before and after the coarse correction.
5. *The coarse correction* (Sec. 3.1). $\tilde{A}_R^{-1}$ factorizes into $C_e\,\pi D^{-1}\iota\,\pi(D_{\mathrm{PV}}^\dagger)^{-1}\iota\,C_{\mathrm{PV},e}^\dagger$; each inverse is replaced by a solve on the coarse lattice, the Pauli–Villars factor by CG on a regularized normal equation and the quark factor by BiCGStab with further levels (Sec. 4).
6. *The coarse space* (Secs. 3.2–3.5 and 5.1). The setup finds vectors y_j near the low modes of $\tilde{A}_R^\dagger\tilde{A}_R$ by a Chebyshev filter and a bootstrap, lifts each to s_j and $\hat{t}_j$ so that the correction is exact on their span (Sec. 3.4), and builds $\mathcal{P}$ and the coarse operators from them.
7. *Solves with $D^\dagger$* (Sec. 8) reuse the same setup through $\Gamma_5 D^{-1}\Gamma_5$.
8. *Why it converges* (Sec. 9): the spectrum and pseudospectra of $\tilde{A}\mathcal{M}$.

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[1] `docs/mg_background/MDWF_MULTIGRID_NOTE.tex` in the RutticeQCD repository.
[2] `docs/mg_background/MG_HMC_WALKTHROUGH.tex`.

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Symbols

The symbols in the order they appear, with the section that defines them. Where the note writes a symbol differently, its name is in the last column.

symbol	meaning	defined	note
$P_\pm$, K, W, M_5	chiral projectors, Wilson hop, Wilson operator of mass $-M_5$	1.1	
b, c, m, L_s, s	Möbius parameters, quark mass, fifth extent and index	1.3	
$D_\pm$, D	$1+bW$, $cW-1$; the domain-wall operator	1.3	
Y, C, $d_\pm$	fifth-dimension hop; site-diagonal part $d_+ + d_- Y$	1.4	
D_{PV}, C_{PV}	D, C at $m=1$	1.4	
R, Γ_5	reflection $s \to L_s-1-s$; $\gamma_5 R$	1.5	
K_D	$D_-^{-1}D_+ - Y$, Γ_5-Hermitian	1.5	
A', X	$(D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$; $X = D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1}$	1.7	
Q, α	Neff's rescaling and its parameter	1.8	
C_e, C_o, D_{eo}, D_{oe}	even-odd blocks of D	2.1	
S	Schur complement of D on the even sites (S_{PV}: of D_{PV})	2.2	
ι, π	embed an even field; take the even part	2.3	
$\tilde{A}_R$, $\tilde{A}_L$	right and left preconditioned forms	2.4	
$\mathcal{K}$, $\mathcal{K}^{(\alpha)}$	left Schur factor $C_e^{-1}S$, and with α	2.4	
$\mathcal{P}$, $\mathcal{D}_{\mathrm{c}}$, $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$	prolongation; coarse operators $\mathcal{P}^\dagger D \mathcal{P}$, $\mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$	3.1	
O_R, O_M, O_L, e	site-local factors of the correction; the correction	3.1	
n_v, Σ	test vectors ($2n_v$ components per coarse site); their chirality signs	3.2	
y_j, s, $\hat{t}$, a	setup vectors; their two lifts; $a = \pi D s$	3.4	
$\phi_{X,k}$, $A(X)$, $B_\mu^\pm(X)$	basis vector on block X; diagonal and link blocks of $\mathcal{D}_{\mathrm{c}}$	3.5	
τ	shift of the regularized Pauli–Villars solve	4	
$\mathcal{D}_\ell$, B_ℓ, $\mathcal{P}_\ell$	operator, diagonal blocks, prolongation of level ℓ	4	Q_ℓ
H_n, $\mathcal{Y}$, ℓ, T_d	$\tilde{A}_R^\dagger \tilde{A}_R$; filter block; interval map; Chebyshev polynomial	5.1	X for $\mathcal{Y}$
$\mathcal{M}$, ν, κ_s	cycle; smoother steps; smoother interval ratio	6	
$\lvert z\rangle$, $\lvert q\rangle$, γ_j, β	GCR direction, $\tilde{A}\lvert z\rangle$, orthogonalization, step length	7	
N	$c[Y,W]D^{-1}$, the error of $\Gamma_5 D^{-1}\Gamma_5$ for $D^\dagger$	8.1	
$\mathcal{A}$, $W(\mathcal{A})$, κ_λ	$\tilde{A}\mathcal{M}$; field of values; eigenvalue condition number	9	

1 The operator and Pauli–Villars preconditioning (note Sec. II.B)

1.1 Ingredients

The field ψ lives on the four-dimensional lattice and carries a fifth index $s = 0, \dots, L_s - 1$; each ψ_s is a Dirac field. The chiral projectors are

$$P_\pm = \frac{1 \pm \gamma_5}{2}, \qquad P_\pm^2 = P_\pm, \qquad P_+ P_- = 0, \qquad P_+ + P_- = 1. \tag{1.1}$$

The Wilson hopping term and the Wilson operator of mass $-M_5$ are

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1-\gamma_\mu) U_\mu(x) \psi(x+\hat{\mu}) + (1+\gamma_\mu) U_\mu^\dagger(x-\hat{\mu}) \psi(x-\hat{\mu}) \Big],$$
$$W = (4 - M_5) - \tfrac{1}{2} K. \tag{1.2}$$

W acts on every slice s alike and does not mix slices.

1.2 The free spectrum of W

The spectrum of W on the free field ($U_\mu = 1$) is needed twice: for the spectra of the note (Sec. II.A) and for D_- (Sec. 8.3). Take a plane wave $\psi(x) = e^{ip\cdot x}u$ with a constant spinor u:

$$\begin{aligned}(K\psi)(x) &= \sum_\mu \Big[(1-\gamma_\mu)e^{ip_\mu} + (1+\gamma_\mu)e^{-ip_\mu}\Big]\psi(x)\\ &= \sum_\mu \Big[(e^{ip_\mu}+e^{-ip_\mu}) - \gamma_\mu(e^{ip_\mu}-e^{-ip_\mu})\Big]\psi(x)\\ &= \sum_\mu \Big[2\cos p_\mu - 2i\gamma_\mu \sin p_\mu\Big]\psi(x),\\ W\psi &= \Big[(4-M_5) - \sum_\mu \cos p_\mu + i\sum_\mu \gamma_\mu \sin p_\mu\Big]\psi\\ &= \Big[-M_5 + \sum_\mu (1-\cos p_\mu) + i\sum_\mu \gamma_\mu \sin p_\mu\Big]\psi. \end{aligned} \tag{1.3}$$

The matrix $G = \sum_\mu \gamma_\mu \sin p_\mu$ is Hermitian, and since the γ_μ anticommute,

$$G^2 = \sum_{\mu,\nu}\gamma_\mu\gamma_\nu \sin p_\mu \sin p_\nu = \sum_\mu \gamma_\mu^2 \sin^2 p_\mu + \sum_{\mu<\nu}\{\gamma_\mu,\gamma_\nu\}\sin p_\mu \sin p_\nu = \sum_\mu \sin^2 p_\mu, \tag{1.4}$$

so its eigenvalues are $\pm(\sum_\mu \sin^2 p_\mu)^{1/2}$. The eigenvalues of W are therefore

$$w(p) = -M_5 + \sum_\mu (1-\cos p_\mu) \pm i\Big(\sum_\mu \sin^2 p_\mu\Big)^{1/2}. \tag{1.5}$$

Their real parts run from $-M_5$ ($p=0$) to $8-M_5$ ($p_\mu = \pi$). A mode bound to the walls exists only where the real part is negative. At the corner with n of the four p_μ equal to π and the others 0 the real part is $2n - M_5$, so for $0 < M_5 < 2$ only the region around $p = 0$ qualifies: the walls carry one light quark and no doublers.

1.3 The operator in Brower–Neff–Orginos form (note Sec. II.B.1)

With b, c the Möbius parameters, define

$$D_+ = 1 + bW, \qquad D_- = cW - 1. \tag{1.6}$$

The operator at quark mass m acts slice by slice as

$$(D\psi)_s = D_+\psi_s + D_-\big(P_-\psi_{s+1} + P_+\psi_{s-1}\big), \qquad \psi_{L_s} \equiv -m\psi_0, \qquad \psi_{-1} \equiv -m\psi_{L_s-1}. \tag{1.7}$$

For $L_s = 4$ the matrix in s is

$$D = \begin{pmatrix} D_+ & D_-P_- & 0 & -mD_-P_+ \\ D_-P_+ & D_+ & D_-P_- & 0 \\ 0 & D_-P_+ & D_+ & D_-P_- \\ -mD_-P_- & 0 & D_-P_+ & D_+ \end{pmatrix}. \tag{1.8}$$

The physical quark field is $q = P_-\psi_0 + P_+\psi_{L_s-1}$; the runs use $b = 1.5$, $c = 0.5$, $L_s = 8$.

1.4 Splitting off the fifth-dimension hop

Define the hop in the fifth dimension, with a minus sign,

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1} \qquad \text{(same wrap-around)}. \tag{1.9}$$

Comparing with Eq. (1.7), the off-diagonal part of D is $-D_-Y$, with D_- on the *left*:

$$D = D_+ - D_-Y. \tag{1.10}$$

Now insert $D_\pm$ and then W, one substitution per line:

$$\begin{aligned} D &= (1+bW) - (cW-1)Y \\ &= 1 + bW - cWY + Y \\ &= 1 + Y + W\,(b - cY) \\ &= 1 + Y + \Big[(4-M_5) - \tfrac{1}{2}K\Big](b-cY) \\ &= \underbrace{\big[1+b(4-M_5)\big]}_{d_+} + \underbrace{\big[1-c(4-M_5)\big]}_{d_-} Y - \tfrac{1}{2}K\,(b-cY) \\ &= C - \tfrac{1}{2}K\,(b-cY), \qquad C \equiv d_+ + d_-Y. \end{aligned} \tag{1.11}$$

C contains no four-dimensional hop: it acts within one site, as an $L_s \times L_s$ matrix in s. All the gauge field sits in $-\frac{1}{2}K(b-cY)$. The Pauli–Villars operator is the same operator at unit mass,

$$D_{\rm PV} = D\big|_{m=1}, \qquad C_{\rm PV} = C\big|_{m=1}. \tag{1.12}$$

1.5 The Γ_5-transformation (note Sec. II.B.2)

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$. Domain-wall fermions use $\Gamma_5 = \gamma_5 R$, $(R\psi)_s = \psi_{L_s-1-s}$. We derive how each block transforms.

K. Read off the matrix elements of K between neighbors:

$$K_{x,x+\hat\mu} = (1-\gamma_\mu)U_\mu(x), \qquad K_{x+\hat\mu,x} = (1+\gamma_\mu)U_\mu^\dagger(x). \tag{1.13}$$

The adjoint swaps the indices and conjugates each block (spin and color commute, and the γ_μ are Hermitian):

$$\begin{aligned} (K^\dagger)_{x,x+\hat\mu} &= \big(K_{x+\hat\mu,x}\big)^\dagger = U_\mu(x)\,(1+\gamma_\mu) = (1+\gamma_\mu)U_\mu(x), \\ (K^\dagger)_{x+\hat\mu,x} &= \big(K_{x,x+\hat\mu}\big)^\dagger = (1-\gamma_\mu)U_\mu^\dagger(x). \end{aligned} \tag{1.14}$$

So $K^\dagger$ is K with $\gamma_\mu \to -\gamma_\mu$. Since $\gamma_5\gamma_\mu\gamma_5 = -\gamma_\mu$,

$$\gamma_5 K \gamma_5 = K^\dagger, \qquad \gamma_5 W \gamma_5 = W^\dagger. \tag{1.15}$$

W acts on every slice alike, so it commutes with R: $\Gamma_5 W \Gamma_5 = W^\dagger$, and the same for $D_\pm$ = functions of W: $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.

Y. Write Y as a matrix in s ($L_s = 4$), from $(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1}$ and the wrap-around $\psi_{L_s} = -m\psi_0$, $\psi_{-1} = -m\psi_{L_s-1}$:

$$Y = \begin{pmatrix} 0 & -P_- & 0 & mP_+ \\ -P_+ & 0 & -P_- & 0 \\ 0 & -P_+ & 0 & -P_- \\ mP_- & 0 & -P_+ & 0 \end{pmatrix}. \tag{1.16}$$

γ_5 commutes with $P_\pm$, so $\Gamma_5 Y \Gamma_5 = RYR$, whose entry (s,s') is $Y_{L_s-1-s,\,L_s-1-s'}$: the matrix rotated by 180°. $Y^\dagger$ has the entry $(Y_{s',s})^\dagger$, the transpose with each $P_\pm$ unchanged ($P_\pm^\dagger = P_\pm$, m real). Both give

$$RYR = Y^\dagger = \begin{pmatrix} 0 & -P_+ & 0 & mP_- \\ -P_- & 0 & -P_+ & 0 \\ 0 & -P_- & 0 & -P_+ \\ mP_+ & 0 & -P_- & 0 \end{pmatrix}, \tag{1.17}$$

so $\Gamma_5 Y \Gamma_5 = Y^\dagger$.

W and Y do not commute. K contains γ_μ, and $\gamma_\mu P_+ = P_-\gamma_\mu$, so a hop of K exchanges the two projectors of Y: $[Y, W] \neq 0$. This one fact makes the Möbius case different from Shamir's.

Consequence: $\Gamma_5 D \Gamma_5 \neq D^\dagger$. From the three rules above and $D = D_+ - D_- Y$,

$$\begin{aligned} \Gamma_5 D \Gamma_5 &= \Gamma_5 D_+ \Gamma_5 - \Gamma_5 D_- \Gamma_5 \, \Gamma_5 Y \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \\ D^\dagger &= D_+^\dagger - (D_- Y)^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger. \end{aligned} \tag{1.18}$$

The order of the last product differs. Subtracting,

$$\begin{aligned} \Gamma_5 D \Gamma_5 - D^\dagger &= Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger \\ &= Y^\dagger (cW^\dagger - 1) - (cW^\dagger - 1) Y^\dagger \\ &= c\,(Y^\dagger W^\dagger - W^\dagger Y^\dagger) = c\,[Y^\dagger, W^\dagger], \end{aligned} \tag{1.19}$$

which is of order c (zero for Shamir, 0.15 relative on a 12^4 stout configuration at $c = 0.5$).

The exact relation. D_+ and D_- are functions of W, so they commute. Factor D_- out to the left (formally):

$$D = D_- K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{1.20}$$

K_D is Γ_5-Hermitian:

$$\begin{aligned} \Gamma_5 K_D \Gamma_5 &= \Gamma_5 D_-^{-1} \Gamma_5 \, \Gamma_5 D_+ \Gamma_5 - \Gamma_5 Y \Gamma_5 \\ &= (D_-^\dagger)^{-1} D_+^\dagger - Y^\dagger \\ &= D_+^\dagger (D_-^\dagger)^{-1} - Y^\dagger \qquad (D_\pm^\dagger \text{ commute}) \\ &= (D_-^{-1} D_+)^\dagger - Y^\dagger = K_D^\dagger. \end{aligned} \tag{1.21}$$

So

$$\begin{aligned} D^\dagger &= K_D^\dagger D_-^\dagger \\ &= \Gamma_5 K_D \Gamma_5 \, D_-^\dagger \\ &= \Gamma_5 D_-^{-1} D \, \Gamma_5 \, D_-^\dagger, \end{aligned} \tag{1.22}$$

and multiplying by $D_- \Gamma_5$ from the left,

$$\boxed{D_- \, \Gamma_5 \, D^\dagger = D \, \Gamma_5 \, D_-^\dagger} \tag{1.23}$$

with no inverse left. D_- itself cannot be inverted in practice: $D_- = c(W - 1/c)$ is c times a Wilson operator of mass $-M_5 - 1/c$ (-3 for $M_5 = 1$, $c = 0.5$), whose spectrum surrounds the origin (Sec. 8.3) (GCR stalls at $|r|^2/|b|^2 \approx 0.74$).

1.6 The site-diagonal part C and its inverse (note Sec. II.B.3)

Where C^{-1} is used. C^{-1} and $(C^\dagger)^{-1}$ enter almost every step of the method:

- the Pauli–Villars preconditioned operator $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$ and the substitution $|\psi\rangle = C^{-1}|y\rangle$ (Sec. 1.7);
- the Schur complement $S = C_e - D_{eo} C_o^{-1} D_{oe}$ (Eq. (2.3)), so every application of S contains one C_o^{-1}; one application of $\tilde{A}_R = (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger (S C_e^{-1})$ (Sec. 2.4) contains four site inversions, C_o^{-1}, C_e^{-1}, $(C_{\mathrm{PV},o}^\dagger)^{-1}$ in $S_{\mathrm{PV}}^\dagger$ and $(C_{\mathrm{PV},e}^{-1})^\dagger$;

- the reconstruction $|\psi_e\rangle = C_e^{-1}|y\rangle$, $|\psi_o\rangle = C_o^{-1}(|\eta_o\rangle - D_{oe}|\psi_e\rangle)$;
- the two lifts $|s\rangle$ and $|\hat{t}\rangle$ of the prolongation, whose odd parts are built with C_o^{-1} and with $(C_{\mathrm{PV},o}^{\dagger})^{-1}$ (Sec. 3.4);
- the solve with $D^{\dagger}$, through $(C_e^{\dagger})^{-1}$ and $(C_o^{\dagger})^{-1}$ (Sec. 8);
- in HMC, $\det C_e \det C_o$ is a constant and drops out of the probability of the links.

Why it must be exact and cheap. If C^{-1} needed an iterative solve, each application of S would contain an inner solve, and the inner tolerance would enter every Krylov step and every cycle. The derivation below shows that C^{-1} has a closed form: it is exact, costs $O(L_s)$ per site, is the same matrix on every site, and does not depend on the gauge field. These four properties are what the uses above assume.

The blocks. Y maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$, so C splits into a block on $\psi^+ = P_+\psi$ and one on $\psi^- = P_-\psi$. On ψ^+, Y brings ψ^+_{s-1} to slice s with the factor -1, and across the wall the factor is $-(-m) = +m$:

$$(C\psi)^+_s = \begin{cases} d_+\psi^+_0 + m\, d_-\psi^+_{L_s-1}, & s = 0, \\ d_+\psi^+_s - d_-\psi^+_{s-1}, & s \geq 1. \end{cases} \tag{1.24}$$

To solve $C^+x = r$, take the rows $s \geq 1$ first:

$$d_+x_s - d_-x_{s-1} = r_s \quad \Longrightarrow \quad x_s = \frac{r_s + d_-x_{s-1}}{d_+}. \tag{1.25}$$

Each x_s is then affine in the unknown x_0. With $q = d_-/d_+$ we claim

$$x_s = p_s + q^s x_0, \qquad p_0 = 0, \qquad p_s = \frac{r_s + d_-p_{s-1}}{d_+}. \tag{1.26}$$

Check by induction:

$$\begin{aligned} x_s &= \frac{r_s + d_-x_{s-1}}{d_+} \\ &= \frac{r_s + d_-(p_{s-1} + q^{s-1}x_0)}{d_+} \\ &= \frac{r_s + d_-p_{s-1}}{d_+} + \frac{d_-}{d_+}q^{s-1}x_0 \\ &= p_s + q^s x_0. \end{aligned} \tag{1.27}$$

The remaining row $s = 0$ fixes x_0:

$$\begin{aligned} d_+x_0 + m\, d_-x_{L_s-1} &= r_0 \\ d_+x_0 + m\, d_-(p_{L_s-1} + q^{L_s-1}x_0) &= r_0 \\ (d_+ + m\, d_-q^{L_s-1})x_0 &= r_0 - m\, d_-p_{L_s-1} \\ x_0 &= \frac{r_0 - m\, d_-p_{L_s-1}}{d_+ + m\, d_-q^{L_s-1}}. \end{aligned} \tag{1.28}$$

The block on ψ^- is the same with $s \to L_s - 1 - s$. So C^{-1} costs $O(L_s)$ per site, is exact, and does not depend on the gauge field. $C^{\dagger}$ has the transposed blocks and is inverted the same way from the other end.

1.7 The Pauli–Villars preconditioned operator (note Sec. II.B.4)

Kanamori, Chen and Matsufuru multiply $D|\psi\rangle = |\eta\rangle$ by $(D_{\rm PV}C_{\rm PV}^{-1})^\dagger$ from the left and set $|\psi\rangle = C^{-1}|y\rangle$:

$$\begin{aligned} D|\psi\rangle &= |\eta\rangle \\ DC^{-1}|y\rangle &= |\eta\rangle \\ (D_{\rm PV}C_{\rm PV}^{-1})^\dagger DC^{-1}|y\rangle &= (D_{\rm PV}C_{\rm PV}^{-1})^\dagger|\eta\rangle \\ A'|y\rangle &= (D_{\rm PV}C_{\rm PV}^{-1})^\dagger|\eta\rangle, \qquad A' \equiv (D_{\rm PV}C_{\rm PV}^{-1})^\dagger DC^{-1}. \end{aligned} \tag{1.29}$$

At $m = 1$ it is Hermitian positive. Then $D = D_{\rm PV}$ and $C = C_{\rm PV}$, so with $X = D_{\rm PV}C_{\rm PV}^{-1}$,

$$A' = X^\dagger X, \qquad \langle z|A'|z\rangle = \big\|X|z\rangle\big\|^2 > 0 \;\; (|z\rangle \neq 0,\; X \text{ invertible}). \tag{1.30}$$

At $m < 1$ it is a perturbation of that. m enters D only in the two corners of the matrix in s ($-mD_-P_+$ and $-mD_-P_-$), and C only in its corner $m\,d_-$:

$$D - D_{\rm PV} = (1-m)\,D_-\,\big(P_+ \text{ in corner } (0, L_s-1) + P_- \text{ in corner } (L_s-1, 0)\big), \tag{1.31}$$

a term localized at the walls. So A' is $X^\dagger X$ plus a wall term; its eigenvalues stay in the right half-plane (note Fig. 2), which is what a Krylov method and a polynomial smoother need.

1.8 Neff's α (note Sec. II.B.6)

Neff's operator is $D^{(\alpha)} = DQ$ with the site-local

$$Q = \mathrm{diag}_s\,\big(P_- + \alpha P_+,\; \alpha,\; \ldots,\; \alpha,\; P_+ + \alpha P_-\big), \tag{1.32}$$

one entry per slice $s = 0, \ldots, L_s - 1$, each acting on the spin. Q is Hermitian and Q^{-1} is the same with $\alpha \to 1/\alpha$. If $D^{(\alpha)}|\psi^{(\alpha)}\rangle = |\eta\rangle$, then

$$|\eta\rangle = D^{(\alpha)}|\psi^{(\alpha)}\rangle = DQ|\psi^{(\alpha)}\rangle \quad\Longrightarrow\quad |\psi\rangle = Q|\psi^{(\alpha)}\rangle \text{ solves } D|\psi\rangle = |\eta\rangle. \tag{1.33}$$

Q leaves $P_-\psi_0$ and $P_+\psi_{L_s-1}$ alone, so the quark propagator is unchanged.

2 Even-odd forms (note Sec. II.C)

2.1 Blocks

Order the sites even first. K connects nearest neighbors, which have opposite parity, so in Eq. (1.11) the diagonal blocks come from C alone:

$$D = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}, \qquad D_{pq} = -\tfrac{1}{2}K_{pq}(b - cY) \quad (p \neq q). \tag{2.1}$$

C_e and C_o are the same matrix C on the even and the odd sites.

2.2 The Schur complement

Solve $D|\psi\rangle = |\eta\rangle$ block by block, with $|\psi\rangle = (|\psi_e\rangle, |\psi_o\rangle)$ and the same for $|\eta\rangle$. The odd rows:

$$\begin{aligned} D_{oe}|\psi_e\rangle + C_o|\psi_o\rangle &= |\eta_o\rangle \\ |\psi_o\rangle &= C_o^{-1}\big(|\eta_o\rangle - D_{oe}|\psi_e\rangle\big). \end{aligned} \tag{2.2}$$

Insert into the even rows:

$$\begin{aligned} C_e|\psi_e\rangle + D_{eo}|\psi_o\rangle &= |\eta_e\rangle \\ C_e|\psi_e\rangle + D_{eo}C_o^{-1}(|\eta_o\rangle - D_{oe}|\psi_e\rangle) &= |\eta_e\rangle \\ (C_e - D_{eo}C_o^{-1}D_{oe})|\psi_e\rangle &= |\eta_e\rangle - D_{eo}C_o^{-1}|\eta_o\rangle \\ S|\psi_e\rangle &= |\eta_e'\rangle, \qquad S \equiv C_e - D_{eo}C_o^{-1}D_{oe}, \quad |\eta_e'\rangle \equiv |\eta_e\rangle - D_{eo}C_o^{-1}|\eta_o\rangle. \end{aligned} \tag{2.3}$$

2.3 $S^{-1} = \pi D^{-1}\iota$

Let ι put an even-site field on the whole lattice (zero on the odd sites) and π take the even part:

$$\iota|y\rangle = \begin{pmatrix} |y\rangle \\ 0 \end{pmatrix}, \qquad \pi \begin{pmatrix} |\psi_e\rangle \\ |\psi_o\rangle \end{pmatrix} = |\psi_e\rangle. \tag{2.4}$$

For $|\eta\rangle = \iota|\eta_e\rangle$ we have $|\eta_o\rangle = 0$, so $|\eta_e'\rangle = |\eta_e\rangle$ and Eq. (2.3) reads $S|\psi_e\rangle = |\eta_e\rangle$. Hence

$$S^{-1}|\eta_e\rangle = |\psi_e\rangle = \pi D^{-1}\iota|\eta_e\rangle \quad \Longrightarrow \quad S^{-1} = \pi D^{-1}\iota. \tag{2.5}$$

The same for $D^\dagger$: its block form is $\begin{pmatrix} C_e^\dagger & D_{oe}^\dagger \\ D_{eo}^\dagger & C_o^\dagger \end{pmatrix}$, and elimination gives $C_e^\dagger - D_{oe}^\dagger (C_o^\dagger)^{-1} D_{eo}^\dagger = S^\dagger$, so $S^{-\dagger} = \pi D^{-\dagger}\iota$.

2.4 The Pauli–Villars preconditioned forms

S is not suitable for a Krylov method: its spectrum surrounds the origin. Following Kanamori, Chen and Matsufuru we multiply from the left by the adjoint of the Pauli–Villars factor and divide the site part out. Two placements of C_e give two forms:

$$\begin{aligned} \tilde{A}_R &= (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \, (S C_e^{-1}), \\ \tilde{A}_L &= (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger \, (C_e^{-1} S). \end{aligned} \tag{2.6}$$

For $m = 1$ both become $X^\dagger X$, Hermitian positive; for $m < 1$ they are perturbations of it, with all eigenvalues in the right half-plane.

Right form. Start from $S|\psi_e\rangle = |\eta_e'\rangle$ and substitute $|\psi_e\rangle = C_e^{-1}|y\rangle$:

$$\begin{aligned} S C_e^{-1}|y\rangle &= |\eta_e'\rangle \\ (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger S C_e^{-1}|y\rangle &= (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger |\eta_e'\rangle \\ \tilde{A}_R|y\rangle &= |\rho_R\rangle, \qquad |\rho_R\rangle \equiv (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger |\eta_e'\rangle, \\ |\psi_e\rangle &= C_e^{-1}|y\rangle, \qquad |\psi_o\rangle = C_o^{-1}(|\eta_o\rangle - D_{oe}|\psi_e\rangle). \end{aligned} \tag{2.7}$$

Left form with α in D only. For DQ the blocks are C_eQ_e, $D_{eo}Q_o$, $D_{oe}Q_e$, C_oQ_o, and the Schur complement is

$$\begin{aligned} S^{(\alpha)} &= C_eQ_e - D_{eo}Q_o(C_oQ_o)^{-1}D_{oe}Q_e \\ &= C_eQ_e - D_{eo}Q_oQ_o^{-1}C_o^{-1}D_{oe}Q_e \\ &= (C_e - D_{eo}C_o^{-1}D_{oe})Q_e \\ &= SQ_e. \end{aligned} \tag{2.8}$$

So the left Schur factor of DQ is

$$\mathcal{K}^{(\alpha)} = (C_e Q_e)^{-1} S Q_e = Q_e^{-1} C_e^{-1} S Q_e = Q_e^{-1} \mathcal{K} Q_e, \qquad \mathcal{K} \equiv C_e^{-1} S, \tag{2.9}$$

and the left form becomes $\tilde{A}_L^{(\alpha)} = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger Q_e^{-1} (C_e^{-1} S) Q_e$. In the right form the two Q_e cancel: $S Q_e (C_e Q_e)^{-1} = S C_e^{-1}$, so $\tilde{A}_R$ does not see α.

3 The factorized coarse-grid correction and the prolongation (note Sec. III.A)

3.1 The factorized coarse correction

The inverse of the right form. Invert the product one factor at a time:

$$\begin{aligned}
\tilde{A}_R^{-1} &= \left[(S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger (S C_e^{-1}) \right]^{-1} \\
&= (S C_e^{-1})^{-1} \left[(S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \right]^{-1} \\
&= C_e S^{-1} \left[(C_{\mathrm{PV},e}^{-1})^\dagger S_{\mathrm{PV}}^\dagger \right]^{-1} \\
&= C_e S^{-1} \, (S_{\mathrm{PV}}^\dagger)^{-1} C_{\mathrm{PV},e}^\dagger \\
&= C_e \; \pi D^{-1} \iota \;\; \pi (D_{\mathrm{PV}}^\dagger)^{-1} \iota \;\; C_{\mathrm{PV},e}^\dagger,
\end{aligned} \tag{3.1}$$

where Eq. (2.5) was used twice. The inverse is a product of the inverses of two *one-hop* operators on the whole lattice, D and $D_{\mathrm{PV}}^\dagger$, and site-local factors.

Galerkin replacement. With a prolongation $\mathcal{P}$ (coarse to whole lattice), replace each inverse by its Galerkin approximation:

$$X^{-1} \approx \mathcal{P} (\mathcal{P}^\dagger X \mathcal{P})^{-1} \mathcal{P}^\dagger. \tag{3.2}$$

Both coarse operators $\mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}$ are nearest-neighbor on the coarse lattice, because D is one hop and every aggregate contains the whole fifth dimension.

Why the Galerkin approximation is exact on $\operatorname{range} \mathcal{P}$**.** For any X with $\mathcal{P}^\dagger X \mathcal{P}$ invertible and any coarse vector $|c\rangle$,

$$\mathcal{P} (\mathcal{P}^\dagger X \mathcal{P})^{-1} \mathcal{P}^\dagger X \, \mathcal{P} |c\rangle = \mathcal{P} (\mathcal{P}^\dagger X \mathcal{P})^{-1} (\mathcal{P}^\dagger X \mathcal{P}) |c\rangle = \mathcal{P} |c\rangle. \tag{3.3}$$

So the approximation of X^{-1} inverts X exactly on vectors $X\mathcal{P}|c\rangle$. This is used in the exactness proof below.

The size of the coarse stencil. An operator that reaches k hops on the fine lattice connects a fine site to sites at most k hops away. With blocks at least two sites wide, two coarse sites can then be coupled only if their coordinates differ by at most 1 in at most k of the four directions. Counting the coarse neighbors that differ in exactly j directions ($\binom{4}{j}$ choices of directions, 2^j signs):

$$\begin{aligned}
k = 1: &\quad 1 + \tbinom{4}{1} 2 = 1 + 8 = 9, \\
k = 2: &\quad 1 + 8 + \tbinom{4}{2} 2^2 = 1 + 8 + 24 = 33, \\
k \geq 4: &\quad \sum_{j=0}^{4} \tbinom{4}{j} 2^j = 3^4 = 81.
\end{aligned} \tag{3.4}$$

D and D_{PV} are one hop: 9 points each. $\tilde{A}_R$ reaches four hops (S is two hops, and $\tilde{A}_R$ contains S and $S_{\mathrm{PV}}^\dagger$): 81 points. That is why the correction factorizes instead of coarsening $\tilde{A}_R$ directly.

The correction of a residual r (Eq. (3.1) read from the right) is

$$e = O_L\,\pi\mathcal{P}\,\mathcal{D}_c^{-1}\mathcal{P}^\dagger\iota\,O_M\,\pi\mathcal{P}\big(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P}\big)^{-1}\mathcal{P}^\dagger\iota\,O_R\,r \tag{3.5}$$

with the site-local factors

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_eQ_eC_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_eQ_e^2C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

Algorithm 1 lists the steps, and Fig. 1 shows them on the three lattices.

Algorithm 1: Factorized coarse-grid correction

input: residual $|r\rangle$ on the even sites
output: correction $|e\rangle$
require: prolongation $\mathcal{P}$; coarse solvers for $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P}$ and $\mathcal{D}_c$; O_R, O_M, O_L of the table

1 $|w\rangle := \iota\,O_R|r\rangle$ `/* to the whole lattice */`
2 $|w_c\rangle := \mathcal{P}^\dagger|w\rangle$ `/* restriction */`
3 $|u_c\rangle := (\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P})^{-1}|w_c\rangle$ `/* Pauli-Villars coarse solve */`
4 $|u\rangle := \mathcal{P}|u_c\rangle$ `/* prolongation */`
5 $|g\rangle := \iota\,O_M\,\pi|u\rangle$ `/* even part, site factor, whole lattice */`
6 $|g_c\rangle := \mathcal{P}^\dagger|g\rangle$ `/* restriction */`
7 $|v_c\rangle := \mathcal{D}_c^{-1}|g_c\rangle$ `/* quark coarse solve */`
8 $|v\rangle := \mathcal{P}|v_c\rangle$ `/* prolongation */`
9 $|e\rangle := O_L\,\pi|v\rangle$ `/* even part, site factor */`

The coarse operators are always those of the plain D and D_{PV}, so one setup serves both forms and every α.

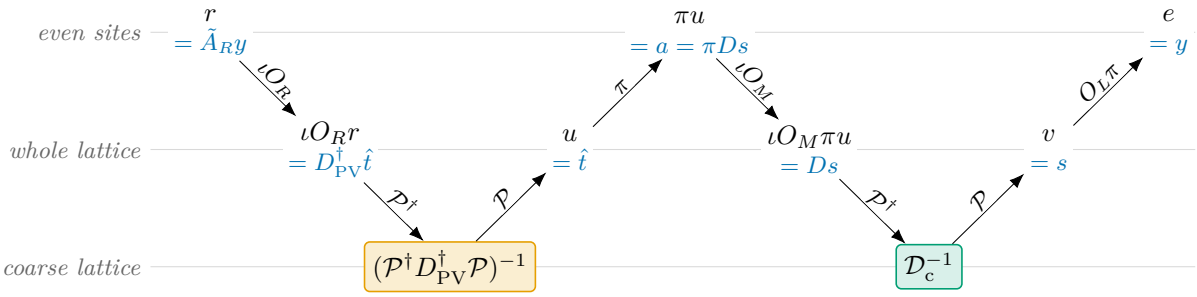


Figure 1: One factorized coarse correction, from r (left) to e (right), on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P})^{-1}\mathcal{P}^\dagger\iota O_R r$ is the result of the first coarse solve, its even part times O_M is the right-hand side of the second, whose result is $v = \mathcal{P}\mathcal{D}_c^{-1}\mathcal{P}^\dagger\iota O_M\pi u$, and $e = O_L\pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range}\mathcal{P}$ (Sec. 3.4; $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift s, and the correction returns y.

The left form with α. For α in D only,

$$\begin{aligned}
\tilde{A}_L^{(\alpha)\,-1} &= \left[(C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger\, Q_e^{-1} C_e^{-1} S Q_e\right]^{-1} \\
&= \left(Q_e^{-1} C_e^{-1} S Q_e\right)^{-1} \left[(C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger\right]^{-1} \\
&= Q_e^{-1} S^{-1} C_e Q_e \left[S_{\mathrm{PV}}^\dagger (C_{\mathrm{PV},e}^{-1})^\dagger\right]^{-1} \\
&= Q_e^{-1} S^{-1} C_e Q_e\, C_{\mathrm{PV},e}^\dagger\, (S_{\mathrm{PV}}^\dagger)^{-1} \\
&= \underbrace{Q_e^{-1}}_{O_L}\, \pi D^{-1} \iota\, \underbrace{C_e Q_e C_{\mathrm{PV},e}^\dagger}_{O_M}\, \pi (D_{\mathrm{PV}}^\dagger)^{-1} \iota\, \underbrace{1}_{O_R}.
\end{aligned} \tag{3.6}$$

Unlike the right form, the Galerkin factor of $D_{\mathrm{PV}}^\dagger$ acts on the plain residual, and $C_{\mathrm{PV},e}^\dagger$ lands between the two coarse solves, in O_M.

3.2 Aggregation and the chirality split

The fine lattice is divided into 4^4 blocks, each with the whole fifth dimension; each block is one coarse site. For a test vector $|v\rangle$ and a block X, restrict it to the block and split it by $\Gamma_5 = \gamma_5 R$, $(R\psi)_s = \psi_{L_s-1-s}$:

$$|X,v\rangle = \begin{cases} |v\rangle, & x \in X, \\ 0, & x \notin X, \end{cases} \qquad |X,v,\pm\rangle = \tfrac{1}{2}(1 \pm \Gamma_5)|X,v\rangle, \qquad \Gamma_5 |X,v,\pm\rangle = \pm |X,v,\pm\rangle. \tag{3.7}$$

Γ_5 acts within a site, so the split commutes with the restriction to X. With n_v test vectors a coarse site has $2n_v$ components, the + parts first. Each column of $\mathcal{P}$ is then an eigenvector of Γ_5, so with $\Sigma = \mathrm{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ per coarse site

$$\Gamma_5 \mathcal{P} = \mathcal{P}\Sigma. \tag{3.8}$$

Half the storage. Write the condition $\Gamma_5 |v_\pm\rangle = \pm |v_\pm\rangle$ out slice by slice, with $v_\pm(s)$ the slice s of $|v_\pm\rangle$:

$$\begin{aligned}
(\Gamma_5 v_\pm)_s &= \gamma_5\, (R v_\pm)_s = \gamma_5\, v_\pm(L_s - 1 - s) \\
\pm v_\pm(s) &= \gamma_5\, v_\pm(L_s - 1 - s) \\
v_\pm(L_s - 1 - s) &= \pm \gamma_5\, v_\pm(s).
\end{aligned} \tag{3.9}$$

So each part is fixed by its slices $s < L_s/2$; $\mathcal{P}$ is stored only there ("folded"), in half precision. Restriction and prolongation unfold on the fly: for the restriction of a fine field $|f\rangle$, with f_s its slice s,

$$\left(\mathcal{P}^\dagger f\right)_{X,k} = \sum_{x \in X} \sum_{s < L_s/2} \mathcal{P}^*_{x,s;k}\, (f_s \pm \gamma_5 f_{L_s-1-s}) \tag{3.10}$$

(the sign that of the chirality of column k), which is what `fold_restrict_tiled_kernel` computes.

3.3 Orthonormalization per block: CholeskyQR2

In each block and chirality the columns $w_1, \ldots, w_n$ ($n = 2\times$ the number of filtered vectors, see below) are made orthonormal. Collect them as the columns of W and write the QR decomposition with a positive diagonal of R:

$$\begin{aligned}
W &= QR \\
W^\dagger W &= R^\dagger Q^\dagger Q R = R^\dagger R \\
G \equiv W^\dagger W &= R^\dagger R \quad \text{(Cholesky)} \\
Q &= W R^{-1}.
\end{aligned} \tag{3.11}$$

So one Gram matrix ($n \times n$), one Cholesky factor and one triangular solve give Q; the factorization is unique, so this Q equals what modified Gram–Schmidt produces. In floating point one pass loses orthogonality like the square of the condition number of W, so the pass is done twice (CholeskyQR2): the second G is close to the identity and repairs the first. Each pass reads W twice, against one sweep per pair of columns for Gram–Schmidt (800 sweeps for $n = 40$). Where a pivot of the first pass falls below 10^{-12} of its diagonal (nearly dependent columns), the code falls back to Gram–Schmidt for those blocks.

3.4 The basis $\{s, \hat{t}\}$ and why the correction is exact

The setup produces vectors $|y_j\rangle$ on the even sites (next section). Each is lifted to the whole lattice in two ways, one for each coarse solve:

$$|s\rangle = \begin{pmatrix} C_e^{-1}|y\rangle \\ -C_o^{-1} D_{oe} C_e^{-1}|y\rangle \end{pmatrix}, \qquad |\hat{t}\rangle = \begin{pmatrix} |a\rangle \\ -(C_{\mathrm{PV},o}^{\dagger})^{-1} D_{\mathrm{PV},eo}^{\dagger}|a\rangle \end{pmatrix}, \qquad |a\rangle = \pi D|s\rangle, \tag{3.12}$$

with the even sites on top, as in the block form of D below.

$D|s\rangle$ vanishes on the odd sites.

$$\begin{aligned} D|s\rangle &= \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix} \begin{pmatrix} C_e^{-1}|y\rangle \\ -C_o^{-1} D_{oe} C_e^{-1}|y\rangle \end{pmatrix} \\ &= \begin{pmatrix} C_e C_e^{-1}|y\rangle - D_{eo} C_o^{-1} D_{oe} C_e^{-1}|y\rangle \\ D_{oe} C_e^{-1}|y\rangle - C_o C_o^{-1} D_{oe} C_e^{-1}|y\rangle \end{pmatrix} \\ &= \begin{pmatrix} (C_e - D_{eo} C_o^{-1} D_{oe}) C_e^{-1}|y\rangle \\ 0 \end{pmatrix} \\ &= \iota\, S C_e^{-1}|y\rangle. \end{aligned} \tag{3.13}$$

In the same way $D_{\mathrm{PV}}^{\dagger}|\hat{t}\rangle = \iota S_{\mathrm{PV}}^{\dagger}|a\rangle$.

Exactness of the right form. Let $|s\rangle, |\hat{t}\rangle \in \operatorname{range} \mathcal{P}$ and solve both coarse systems exactly. The Galerkin map $\mathcal{P}(\mathcal{P}^{\dagger} X \mathcal{P})^{-1} \mathcal{P}^{\dagger} X$ is the identity on $\operatorname{range} \mathcal{P}$. Follow $|r\rangle = \tilde{A}_R|y\rangle$ through Algorithm 1 ($O_R = C_{\mathrm{PV},e}^{\dagger}$, $O_M = 1$, $O_L = C_e$):

$$\begin{aligned} O_R|r\rangle &= C_{\mathrm{PV},e}^{\dagger} (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^{\dagger} S C_e^{-1}|y\rangle \\ &= C_{\mathrm{PV},e}^{\dagger} (C_{\mathrm{PV},e}^{-1})^{\dagger} S_{\mathrm{PV}}^{\dagger} S C_e^{-1}|y\rangle \\ &= S_{\mathrm{PV}}^{\dagger}|a\rangle, \qquad (|a\rangle = S C_e^{-1}|y\rangle = \pi D|s\rangle) \\ \iota O_R|r\rangle &= \iota S_{\mathrm{PV}}^{\dagger}|a\rangle = D_{\mathrm{PV}}^{\dagger}|\hat{t}\rangle \\ \mathcal{P}(\mathcal{P}^{\dagger} D_{\mathrm{PV}}^{\dagger} \mathcal{P})^{-1} \mathcal{P}^{\dagger}\, D_{\mathrm{PV}}^{\dagger}|\hat{t}\rangle &= |\hat{t}\rangle \qquad (|\hat{t}\rangle \in \operatorname{range} \mathcal{P}) \\ \pi|\hat{t}\rangle &= |a\rangle = \pi D|s\rangle \\ \iota O_M \pi|\hat{t}\rangle &= \iota \pi D|s\rangle = D|s\rangle \qquad (D|s\rangle \text{ is zero on the odd sites}) \\ \mathcal{P}\mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^{\dagger} D|s\rangle &= |s\rangle \qquad (|s\rangle \in \operatorname{range} \mathcal{P}) \\ |e\rangle = O_L \pi|s\rangle &= C_e C_e^{-1}|y\rangle = |y\rangle. \end{aligned} \tag{3.14}$$

So the correction inverts $\tilde{A}_R$ exactly on the span of the y_j. In practice $\mathcal{P}$ is in half precision and the coarse solves are inexact ($|r|^2/|b|^2 < 10^{-2}$), and the left form is not exact at all; the cycle converges nonetheless.

Why the left form is not exact. Repeat the argument for $\tilde{A}_L$ with $\alpha = 1$ and $|r\rangle = \tilde{A}_L|y'\rangle$, $|y'\rangle = C_e^{-1}|y\rangle$ ($O_R = 1$, $O_M = C_e C_{\mathrm{PV},e}^{\dagger}$, $O_L = 1$):

$$|r\rangle = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^{\dagger} C_e^{-1} S\, C_e^{-1}|y\rangle = S_{\mathrm{PV}}^{\dagger} (C_{\mathrm{PV},e}^{-1})^{\dagger} C_e^{-1}|a\rangle, \qquad |a\rangle = S C_e^{-1}|y\rangle,$$
$$\iota O_R|r\rangle = \iota S_{\mathrm{PV}}^{\dagger}|u\rangle, \qquad |u\rangle \equiv (C_{\mathrm{PV},e}^{\dagger})^{-1} C_e^{-1}|a\rangle. \tag{3.15}$$

The first solve returns the Pauli–Villars lift of $|u\rangle$, not of $|a\rangle$. $|u\rangle$ differs from $|a\rangle$ by the site-local matrix $(C_{\mathrm{PV},e}^{\dagger})^{-1} C_e^{-1}$, which mixes the slices s and the spin, so its lift is in general not in the span of the $\hat{t}_j$ made from a. The correction is then only approximate. In practice the left form converges within a few iterations of the right form (note Sec. V), so the cycle does not need the exactness.

3.5 The coarse operators by probing

Write $\phi_{X,k}$ for column k ($k = 1, \ldots, 2n_v$) of $\mathcal{P}$ on block X: it equals the orthonormalized vector on the sites of X and vanishes outside X. By the two previous subsections

$$\langle \phi_{X,j} | \phi_{Y,k} \rangle = \delta_{XY} \delta_{jk}, \qquad \mathcal{P} = \sum_{X,k} |\phi_{X,k}\rangle\langle X, k|, \tag{3.16}$$

where $|X, k\rangle$ is the unit vector of component k on coarse site X. The coarse operator $\mathcal{D}_{\mathrm{c}} = \mathcal{P}^{\dagger} D \mathcal{P}$ (the "little Dirac operator" of Lüscher's deflation) and the coarse correction are then

$$(\mathcal{D}_{\mathrm{c}})_{jk}(X, Y) = \langle \phi_{X,j} | D | \phi_{Y,k} \rangle, \qquad \mathcal{P} \mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^{\dagger} = \sum_{X,Y,j,k} |\phi_{X,j}\rangle\, (\mathcal{D}_{\mathrm{c}}^{-1})_{jk}(X, Y)\, \langle \phi_{Y,k}|, \tag{3.17}$$

and in the same way for $D_{\mathrm{PV}}^{\dagger}$. D hops only to nearest neighbors in four dimensions and a block holds the whole fifth dimension, so $(\mathcal{D}_{\mathrm{c}})(X, Y)$ vanishes unless $Y = X$ or $Y = X \pm \hat{\mu}$. Each coarse site X therefore has a $2n_v \times 2n_v$ diagonal block $A(X) = \mathcal{D}_{\mathrm{c}}(X, X)$ and one link block $B_{\mu}^{\pm}(X) = \mathcal{D}_{\mathrm{c}}(X, X \pm \hat{\mu})$ per direction: nine blocks. They are found column by column. For component k, put the unit vector of component k on *every* coarse site and prolong: $|\phi\rangle = \mathcal{P} \sum_Y |Y, k\rangle = \sum_Y |\phi_{Y,k}\rangle$. Split $D = C + H$ (H the hops) and H by leg (direction and sign), and split each leg into the part that stays inside its block and the part that crosses to the neighbor block:

$$\begin{aligned} D|\phi\rangle &= C|\phi\rangle + \sum_{\text{legs } l} H_l |\phi\rangle \\ &= \underbrace{C|\phi\rangle + \sum_{l} H_l^{\mathrm{in}}|\phi\rangle}_{\text{stays in } X} + \sum_{l} \underbrace{H_l^{\mathrm{cross}}|\phi\rangle}_{\text{comes from } X \pm \hat{\mu}} . \end{aligned} \tag{3.18}$$

Restricting to block X,

$$\langle \phi_{X,j} | \Big(C + \sum_{l} H_l^{\mathrm{in}} \Big) |\phi\rangle = A_{jk}(X), \qquad \langle \phi_{X,j} | H_l^{\mathrm{cross}} | \phi \rangle = \big(B_l(X)\big)_{jk}, \tag{3.19}$$

column k of the diagonal block and of the link block of leg l: every coarse site carries the same unit vector, and a crossing hop brings exactly the neighbor's contribution. Doing this for all $2n_v$ components fills all nine blocks (`coarse_blocks`, `hop_split`, `restrict_leg`). Several components are probed at a time, as many as a quarter of the free device memory allows (four on 24^4), so that $\mathcal{P}$ is read once for them.

4 The coarse solvers (note Sec. III.B)

Quark factor. $\mathcal{D}_c|c\rangle = |g\rangle$ by BiCGStab on its even-odd Schur complement, preconditioned by the inverse diagonal blocks. With $\mathcal{D}_c = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice, exactly as in Eq. (2.3),

$$\begin{aligned} (A_{ee} - B_{eo}A_{oo}^{-1}B_{oe})|c_e\rangle &= |g_e\rangle - B_{eo}A_{oo}^{-1}|g_o\rangle, \\ |c_o\rangle &= A_{oo}^{-1}\big(|g_o\rangle - B_{oe}|c_e\rangle\big). \end{aligned} \tag{4.1}$$

Pauli–Villars factor. $X|w\rangle = |g\rangle$ with $X = \mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = (\mathcal{D}_c^{\mathrm{PV}})^\dagger$. The code stores X itself (the adjoint of the probed $\mathcal{D}_c^{\mathrm{PV}}$, `CoarseOp::adjoint`) and a second copy $\mathcal{D}_c^{\mathrm{PV}} = X^\dagger$, and solves the Tikhonov-regularized normal equation by CG on all coarse sites, without a preconditioner:

$$(X^\dagger X + \tau)|w\rangle = X^\dagger|g\rangle, \qquad \tau = 10^{-3}. \tag{4.2}$$

To see what the shift does, write the singular value decomposition $X = \sum_k \sigma_k |u_k\rangle\langle v_k|$ ($\sigma_k > 0$, the $|u_k\rangle$ and the $|v_k\rangle$ orthonormal). The exact solution is $|w\rangle = \sum_k \sigma_k^{-1}\langle u_k|g\rangle\,|v_k\rangle$; the regularized one is

$$|w_\tau\rangle = \sum_k \frac{\sigma_k}{\sigma_k^2 + \tau}\,\langle u_k|g\rangle\,|v_k\rangle = \sum_k \frac{\sigma_k^2}{\sigma_k^2 + \tau}\,\frac{\langle u_k|g\rangle}{\sigma_k}\,|v_k\rangle, \tag{4.3}$$

so each component is multiplied by $\sigma_k^2/(\sigma_k^2 + \tau)$: close to 1 for $\sigma_k \gg \sqrt{\tau}$, close to σ_k^2/τ for $\sigma_k \ll \sqrt{\tau}$. With stout smearing the Galerkin factor has singular values far below those of the fine D_{PV} (note Sec. V.C.3); the exact solve would amplify those directions by $1/\sigma_k$, and the shift caps the amplification at $1/(2\sqrt{\tau})$. CG on $X^\dagger X + \tau$ needs one application of X and one of $X^\dagger$ per iteration. Both factors stop at $|r|^2/|b|^2 < 10^{-2}$ (for the Pauli–Villars factor, the residual of the normal equation).

Levels below. At small masses the coarse solves themselves become expensive (on 24^4 at $m = 0.001$ about 2000 applications of $\mathcal{D}_c$ per correction). The coarse sites are then grouped into superblocks (2, then 3), local vectors are formed from $\mathcal{P}^\dagger|s_j\rangle$, $\mathcal{P}^\dagger|\hat{t}_j\rangle$, and a V-cycle preconditions the coarse solve of the quark factor. Number the levels from the coarse lattice, $\ell = 1$, down; let $\mathcal{D}_\ell$ be the operator on level ℓ, B_ℓ its diagonal blocks, and $\mathcal{P}_{\ell+1}$ the prolongation from level $\ell+1$ to level ℓ ($Q_{\ell+1}$ in the note). The preconditioner M_ℓ on level ℓ is defined from the one below (Algorithm 2).

Algorithm 2: Preconditioner M_ℓ on level ℓ (V-cycle)

input: vector $|r\rangle$ on level ℓ
output: $M_\ell|r\rangle$
require: $\mathcal{D}_\ell$, its diagonal blocks B_ℓ, the prolongation $\mathcal{P}_{\ell+1}$

1 **if** ℓ is the last level **then return** $\mathcal{D}_\ell^{-1}|r\rangle$ `/* dense inverse */`
2 $|z\rangle := \mathcal{P}_{\ell+1}\, M_{\ell+1}\, \mathcal{P}_{\ell+1}^\dagger|r\rangle$ `/* restrict, recurse, prolong */`
3 **return** $|z\rangle + B_\ell^{-1}\big(|r\rangle - \mathcal{D}_\ell|z\rangle\big)$ `/* one block-Jacobi step */`

This brings the coarse cost down to about 50 applications per correction.

5 Setup (note Sec. III.C)

The setup produces the vectors y_j, then $\{s_j, \hat{t}_j\}$, $\mathcal{P}$, the coarse operators and the levels. It always uses the right form.

5.1 Chebyshev-filtered subspace iteration

The y_j should approximate the low right singular vectors of $\tilde{A}_R$, i.e. the low eigenvectors of $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. With $\ell(x) = (2x - (\lambda_{\rm hi} + \lambda_{\rm lo}))/(\lambda_{\rm hi} - \lambda_{\rm lo})$, the polynomial $T_d(\ell(x))$ is bounded by 1 on $[\lambda_{\rm lo}, \lambda_{\rm hi}]$ and grows fast below $\lambda_{\rm lo}$. Collect the n_f vectors as the columns of $\mathcal{Y} = (y_1, \ldots, y_{n_f})$ (X in the note's Algorithm 1). One round is Algorithm 3.

Algorithm 3: One round of the Chebyshev filter

input: $\mathcal{Y} = (|y_1\rangle, \ldots, |y_{n_f}\rangle)$; interval $[\lambda_{\rm lo}, \lambda_{\rm hi}]$
output: filtered and rotated $\mathcal{Y}$; new $\lambda_{\rm lo}$
require: $H_n = \tilde{A}_R^\dagger \tilde{A}_R$; $L = \ell(H_n)$

1 $d := 1.5\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$, kept between 40 and 1000 `/* degree */`
2 $\mathcal{Y}_0 := \mathcal{Y}$; $\mathcal{Y}_1 := L\mathcal{Y}$
3 **for** $k = 1, \ldots, d-1$ **do** $\mathcal{Y}_{k+1} := 2L\,\mathcal{Y}_k - \mathcal{Y}_{k-1}$ **end** `/* three-term recurrence */`
4 $\mathcal{Y} := \mathcal{Y}_d$ `/* =` $T_d(L)\mathcal{Y}$`, d applications of` H_n `*/`
5 orthonormalize $\mathcal{Y}$
6 $G := \mathcal{Y}^\dagger H_n \mathcal{Y} = V\Theta V^\dagger$; $\mathcal{Y} := \mathcal{Y}V$ `/* Rayleigh-Ritz */`
7 $\lambda_{\rm lo} := \theta_{\max}$

G is the $n_f \times n_f$ matrix of H_n in the span of $\mathcal{Y}$; its eigenvalues $\theta_1 \le \cdots \le \theta_{\max}$ (the Ritz values) estimate the n_f lowest eigenvalues of H_n, and $\mathcal{Y}V$ the eigenvectors. Setting $\lambda_{\rm lo} = \theta_{\max}$ makes the next round damp everything above the span already found and amplify what lies below it. Lines 2–3 apply the filter without the coefficients of the polynomial, through $T_{k+1}(L) = 2L\,T_k(L) - T_{k-1}(L)$ with $T_0 = 1$, $T_1 = L$.

How much the filter amplifies, and the degree. For $|t| \ge 1$, $|T_d(t)| = \cosh(d \operatorname{arccosh}|t|)$. A mode with x just below $\lambda_{\rm lo}$ has $\ell(x) = -(1+\delta)$ with small $\delta > 0$, and

$$\operatorname{arccosh}(1+\delta) = \ln\Big(1 + \delta + \sqrt{2\delta + \delta^2}\Big) \approx \sqrt{2\delta},$$
$$|T_d(\ell(x))| \approx \cosh\big(d\sqrt{2\delta}\big). \tag{5.1}$$

The lowest modes have $x \approx 0$, where

$$\ell(0) = -\frac{\lambda_{\rm hi} + \lambda_{\rm lo}}{\lambda_{\rm hi} - \lambda_{\rm lo}} \approx -\Big(1 + 2\frac{\lambda_{\rm lo}}{\lambda_{\rm hi}}\Big), \qquad \delta \approx 2\frac{\lambda_{\rm lo}}{\lambda_{\rm hi}}, \qquad |T_d(\ell(0))| \approx \cosh\Big(2d\sqrt{\lambda_{\rm lo}/\lambda_{\rm hi}}\Big), \tag{5.2}$$

while every mode inside $[\lambda_{\rm lo}, \lambda_{\rm hi}]$ stays below 1. With the code's degree $d = 1.5\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$ the lowest modes gain a factor $\cosh 3 \approx 10$ per round over the damped ones: the degree is chosen for a fixed amplification per round.

In the code the degree is $d = 1.5\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$, between 40 and 1000; the first round uses $\lambda_{\rm lo} = \lambda_{\rm hi}/1000$. *Observation* ($24^4$, $m = 0.001$, 20 vectors): after the first round (degree 48) the largest Ritz value is still $0.34\,\lambda_{\rm hi}$, so the second round has degree 40 and damps only the top two thirds of the spectrum. The filter is weak; the bootstrap does most of the work.

5.2 Bootstrap

Each round is Algorithm 4.

Algorithm 4: One bootstrap round

input: vectors $|y_1\rangle, \ldots, |y_n\rangle$ on the even sites
output: improved $|y_j\rangle$

lift each $|y_j\rangle$ to $|s_j\rangle$ and $|\hat{t}_j\rangle$ /* Sec. 3.4 */
split by chirality, orthonormalize per block (CholeskyQR2): $\mathcal{P}$ /* Secs. 3.2-3.4 */
probe $\mathcal{D}_c$ and $\mathcal{D}_c^{\mathrm{PV}}$; build the levels below /* Secs. 3.5, 4 */
build the cycle $\mathcal{M}$ /* Algorithm 5 */
for each j **do** $|y_j\rangle :=$ four steps of FGMRES on $\tilde{A}_R|x\rangle = |y_j\rangle$ from $|x\rangle = 0$ with $\mathcal{M}$ **end**
orthonormalize the $|y_j\rangle$ through their Gram matrix

To see why this works, write y_j in eigenvectors of the preconditioned operator. The cycle already handles the components inside its coarse space, and FGMRES with it removes them first; what remains, and is therefore amplified relative to the rest, is what the coarse space misses. In HMC the levels below are used from the second round on (in the first, on the filtered vectors, the coarse solves were as fast without them).

5.3 Cost and memory

On 24^4 ($L_s = 8$, 20 filtered vectors, $m = 0.001$; the $24^3 \times 48$ runs of the note use 16, $n = 32$): the filter about 35 s and the bootstrap about 77 s, the setup in all about 113 s. A bootstrap round makes its correction ($\mathcal{P}$ 1.8 s, the two coarse operators about 4.4 s each, the inverse blocks 0.8 s, the top of the smoother 1.3 s; the levels 7 s more in the second round) and then improves the 20 vectors (about 20 s in the first round, 26 s in the second). $\mathcal{P}$ (40 columns, half precision, whole lattice) is the largest object, about 4.7 GB.

6 The Chebyshev smoother and the cycle (note Sec. III.D)

One cycle is Algorithm 5.

Algorithm 5: Multigrid cycle

input: source vector $|b\rangle$ on the even sites
output: $|y\rangle = \mathcal{M}|b\rangle$
require: Chebyshev smoother $p(\tilde{A})$ (Algorithm 6); coarse correction E (Algorithm 1)

$|y\rangle := p(\tilde{A})|b\rangle$ /* pre-smoothing */
$|r\rangle := |b\rangle - \tilde{A}|y\rangle$
$|y\rangle := |y\rangle + E|r\rangle$ /* coarse correction */
$|r\rangle := |b\rangle - \tilde{A}|y\rangle$
$|y\rangle := |y\rangle + p(\tilde{A})|r\rangle$ /* post-smoothing */

Cost of one cycle. Count the applications of $\tilde{A}$: the pre-smoothing (ν Chebyshev steps from $x = 0$) costs $\nu - 1$, the residual after it 1, the residual after the correction 1, the post-smoothing $\nu - 1$:

$$(\nu - 1) + 1 + 1 + (\nu - 1) = 2\nu, \tag{6.1}$$

plus the two coarse solves. One application of $\tilde{A}$ is two Schur factors, each two hops: $\nu = 8$ means 16 applications, 64 applications of D_{eo} or D_{oe}, per cycle. This is why one multigrid iteration costs as much as about 24 iterations of CG on $\mathcal{K}^\dagger\mathcal{K}$ (two hops each with $\mathcal{K}$ and $\mathcal{K}^\dagger$).

6.1 The Chebyshev smoother, derived

The smoother is ν steps of Chebyshev iteration on $\tilde{A}|x\rangle = |b\rangle$ from $|x\rangle = 0$. Choose the interval $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ that it damps (λ_{hi} slightly above the spectral radius from a power iteration, $\lambda_{\mathrm{lo}} = \lambda_{\mathrm{hi}}/\kappa_s$)

and set

$$\theta = \tfrac{1}{2}(\lambda_{\rm hi} + \lambda_{\rm lo}), \qquad \delta = \tfrac{1}{2}(\lambda_{\rm hi} - \lambda_{\rm lo}), \qquad \sigma = \theta/\delta. \tag{6.2}$$

The residual after k steps is $|r_k\rangle = R_k(\tilde{A})|r_0\rangle$ with the scaled Chebyshev polynomial

$$R_k(\lambda) = \frac{T_k\big((\theta-\lambda)/\delta\big)}{T_k(\sigma)}, \qquad R_k(0) = 1, \tag{6.3}$$

which is at most $1/T_k(\sigma)$ in modulus on the interval. With ℓ of Sec. 5.1 built on this interval, $(\theta-\lambda)/\delta = -\ell(\lambda)$ and $\sigma = -\ell(0)$, so $R_k(\lambda) = T_k(\ell(\lambda))/T_k(\ell(0))$: the polynomial of the setup filter, normalized to 1 at the origin. The filter applies it to H_n to amplify the modes below $\lambda_{\rm lo}$; the smoother applies it to $\tilde{A}$ to damp the modes in $[\lambda_{\rm lo}, \lambda_{\rm hi}]$ and leaves the low modes to the coarse correction. Write $t = (\theta-\lambda)/\delta$ and use $T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t)$:

$$\begin{aligned} T_{k+1}(\sigma)R_{k+1} &= 2t\,T_k(\sigma)R_k - T_{k-1}(\sigma)R_{k-1} \\ R_{k+1} &= \rho_k\big[2t\,R_k - \rho_{k-1}R_{k-1}\big], \qquad \rho_k \equiv \frac{T_k(\sigma)}{T_{k+1}(\sigma)}. \end{aligned} \tag{6.4}$$

The same recurrence at $t = \sigma$ gives the ρ's:

$$\begin{aligned} T_{k+1}(\sigma) &= 2\sigma T_k(\sigma) - T_{k-1}(\sigma) \\ \frac{1}{\rho_k} &= 2\sigma - \rho_{k-1}, \qquad \rho_0 = \frac{T_0(\sigma)}{T_1(\sigma)} = \frac{1}{\sigma}. \end{aligned} \tag{6.5}$$

Now split $2t = 2\sigma - 2\lambda/\delta$ and use $2\sigma = 1/\rho_k + \rho_{k-1}$:

$$\begin{aligned} R_{k+1} &= \rho_k\Big[(2\sigma - \tfrac{2\lambda}{\delta})R_k - \rho_{k-1}R_{k-1}\Big] \\ &= \rho_k\Big[(\tfrac{1}{\rho_k} + \rho_{k-1})R_k - \tfrac{2\lambda}{\delta}R_k - \rho_{k-1}R_{k-1}\Big] \\ &= R_k + \rho_k\rho_{k-1}(R_k - R_{k-1}) - \frac{2\rho_k}{\delta}\lambda R_k. \end{aligned} \tag{6.6}$$

Apply to $|r_0\rangle$, with $\lambda \to \tilde{A}$:

$$|r_{k+1}\rangle - |r_k\rangle = \rho_k\rho_{k-1}(|r_k\rangle - |r_{k-1}\rangle) - \frac{2\rho_k}{\delta}\tilde{A}|r_k\rangle. \tag{6.7}$$

Since $|r\rangle = |b\rangle - \tilde{A}|x\rangle$, a step $|x_{k+1}\rangle = |x_k\rangle + |d_k\rangle$ changes the residual by $-\tilde{A}|d_k\rangle$. Matching the two sides,

$$|d_k\rangle = \rho_k\rho_{k-1}\,|d_{k-1}\rangle + \frac{2\rho_k}{\delta}\,|r_k\rangle. \tag{6.8}$$

The first step: $R_1 = T_1(t)/T_1(\sigma) = t/\sigma = 1 - \lambda/\theta$, so $|x_1\rangle = |b\rangle/\theta$ and $|d_0\rangle = |r_0\rangle/\theta$. Algorithm 6 collects these updates, as in the code (`BlasChebyshevStep`): ν steps cost $\nu-1$ applications of $\tilde{A}$, and the polynomial is fixed, so the smoother is a linear map.

Algorithm 6: Chebyshev smoother, ν steps from $|x\rangle = 0$

input: source vector $|b\rangle$; ν; θ, δ, $\sigma = \theta/\delta$ of the interval
output: $|x\rangle = p(\tilde{A})|b\rangle$

$\rho_0 := 1/\sigma$; $\quad |r\rangle := |b\rangle$; $\quad |d\rangle := |r\rangle/\theta$; $\quad |x\rangle := |d\rangle$
for $k = 1, \ldots, \nu-1$ **do**
  $|r\rangle := |r\rangle - \tilde{A}|d\rangle$ `/* the one application of Ã */`
  $\rho_k := 1/(2\sigma - \rho_{k-1})$
  $|d\rangle := \rho_k\rho_{k-1}|d\rangle + (2\rho_k/\delta)\,|r\rangle$
  $|x\rangle := |x\rangle + |d\rangle$
end

The runs use $\nu = 8$, $\kappa_s = 10$ with stout links ($M_5 = 1$) and $\nu = 12$, $\kappa_s = 30$ without ($M_5 = 1.8$).

7 The outer solver: flexible GCR (note Sec. III.E)

The outer iteration solves $\tilde{A}|y\rangle = |\rho\rangle$ ($\tilde{A} = \tilde{A}_R$ or $\tilde{A}_L$) with a right preconditioner $\mathcal{M}$, the multigrid cycle. Because $\mathcal{M}$ contains Krylov solves stopped by a residual test, it changes from one application to the next, so the method must be flexible: it keeps the preconditioned directions $|z\rangle$ themselves.

Algorithm 7: Flexible GCR(k)

input: source vector $|\rho\rangle$; k; tolerance ϵ
output: $|y\rangle$ with $\||\rho\rangle - \tilde{A}|y\rangle\|^2 < \epsilon\,\langle\rho|\rho\rangle$
require: cycle $\mathcal{M}$ (Algorithm 5)

```
1  |y⟩ := 0;   |r⟩ := |ρ⟩
2  while ⟨r|r⟩ ≥ ε ⟨ρ|ρ⟩ do
3      |z⟩ := M|r⟩                                   /* one cycle */
4      |q⟩ := Ã|z⟩
5      for each of the last k kept pairs (|z_j⟩, |q_j⟩) do
6          γ_j := ⟨q_j|q⟩/⟨q_j|q_j⟩;   |q⟩ := |q⟩ − γ_j|q_j⟩;   |z⟩ := |z⟩ − γ_j|z_j⟩
7      end
8      β := ⟨q|r⟩/⟨q|q⟩
9      |y⟩ := |y⟩ + β|z⟩;   |r⟩ := |r⟩ − β|q⟩
10     keep (|z⟩, |q⟩)
11 end
```

Line 6 applies the projector $1 - \sum_j |q_j\rangle\langle q_j|/\langle q_j|q_j\rangle$ to $|q\rangle$, so the kept $|q_j\rangle$ are mutually orthogonal, and the same combination of the $|z_j\rangle$ keeps $\tilde{A}|z\rangle = |q\rangle$ true. The step length β of line 8 (γ and β as in the note's Algorithm 4, which also normalizes each $|q\rangle$) minimizes $\||r\rangle - \beta|q\rangle\|$: it removes from $|r\rangle$ its component along $|q\rangle$, and $|r\rangle = |\rho\rangle - \tilde{A}|y\rangle$ holds exactly after line 9. With $k = 2$ (GCR(2)) the code keeps $k + 1 = 3$ pairs (z, q) and the residual, seven fields on the even sites, at the price of no optimality over the whole Krylov space.

8 Solving with $D^\dagger$ (note Sec. IV)

HMC needs $(D^\dagger D)^{-1}$, so it solves with $D^\dagger$ as often as with D. A second setup for $D^\dagger$ would double the cost. The tool that avoids it is an exact relation between D and $D^\dagger$.

The Γ_5-transformation and the exact relation it rests on were derived in Sec. 1.5.

8.1 $\Gamma_5 D^{-1}\Gamma_5$ as a preconditioner of $D^\dagger$ (note Sec. IV.A)

Multiply $D^\dagger$ from the right by $\Gamma_5 D^{-1}\Gamma_5$ and use the exact relation of Sec. 1.5 in the form $D^\dagger = \Gamma_5 D_-^{-1} D\,\Gamma_5 D_-^\dagger$, together with $\Gamma_5 D_-^\dagger \Gamma_5 = D_-$:

$$\begin{aligned} D^\dagger\,\Gamma_5 D^{-1}\Gamma_5 &= \Gamma_5 D_-^{-1} D\,\Gamma_5 D_-^\dagger\,\Gamma_5\,D^{-1}\Gamma_5 \\ &= \Gamma_5 D_-^{-1} D\,D_-\,D^{-1}\,\Gamma_5 \\ &= \Gamma_5 D_-^{-1}(D_- D + [D, D_-])D^{-1}\Gamma_5 \\ &= \Gamma_5(1 + D_-^{-1}[D, D_-]D^{-1})\Gamma_5. \end{aligned} \tag{8.1}$$

The commutator, with $D = D_+ - D_- Y$ and $[D_+, D_-] = 0$:

$$\begin{aligned} [D, D_-] &= [D_+, D_-] - [D_- Y, D_-] \\ &= -(D_- Y D_- - D_- D_- Y) = -D_-[Y, D_-] \\ &= -D_-[Y, cW - 1] = -c\,D_-[Y, W]. \end{aligned} \tag{8.2}$$

The factor D_- cancels against D_-^{-1}:

$$\boxed{D^\dagger\,\Gamma_5 D^{-1}\Gamma_5 = \Gamma_5\,(1 - N)\,\Gamma_5, \qquad N = c\,[Y, W]\,D^{-1}} \tag{8.3}$$

The commutator written out. Split the hop into its spin-scalar part and its γ part, $K = K_0 - K_\gamma$ with

$$\begin{aligned}(K_0\psi)(x) &= \sum_\mu \big[U_\mu(x)\psi(x+\hat\mu) + U_\mu^\dagger(x-\hat\mu)\psi(x-\hat\mu)\big],\\ (K_\gamma\psi)(x) &= \sum_\mu \gamma_\mu\big[U_\mu(x)\psi(x+\hat\mu) - U_\mu^\dagger(x-\hat\mu)\psi(x-\hat\mu)\big].\end{aligned} \tag{8.4}$$

K_0 commutes with Y. K_γ does not: $\gamma_\mu P_\mp = P_\pm\gamma_\mu$, so $K_\gamma P_\mp = P_\pm K_\gamma$. Slice by slice,

$$\begin{aligned}(YK_\gamma\psi)_s &= -P_- K_\gamma\psi_{s+1} - P_+ K_\gamma\psi_{s-1},\\ (K_\gamma Y\psi)_s &= -P_+ K_\gamma\psi_{s+1} - P_- K_\gamma\psi_{s-1},\\ ([Y,K_\gamma]\psi)_s &= (P_+ - P_-)K_\gamma(\psi_{s+1}-\psi_{s-1}) = \gamma_5 K_\gamma(\psi_{s+1}-\psi_{s-1}),\end{aligned} \tag{8.5}$$

with the wrap-around of Y. The constant part of W commutes with Y, so $[Y,W] = -\frac12[Y,K] = \frac12[Y,K_\gamma]$:

$$\big([Y,W]\psi\big)_s = \tfrac12\,\gamma_5 K_\gamma(\psi_{s+1}-\psi_{s-1}). \tag{8.6}$$

It is a difference in s followed by one hop that flips chirality (γ_5 keeps it, γ_μ flips it). It is bounded, so N is large only through D^{-1}.

N^2 written out. Expand D^{-1} in the right and left eigenvectors of D (assumed diagonalizable), $D|u_k\rangle = \lambda_k|u_k\rangle$, $\langle w_k|D = \lambda_k\langle w_k|$, $\langle w_k|u_l\rangle = \delta_{kl}$:

$$\begin{aligned}D^{-1} &= \sum_k \frac{|u_k\rangle\langle w_k|}{\lambda_k},\\ N &= c\sum_k \frac{[Y,W]|u_k\rangle\,\langle w_k|}{\lambda_k},\\ N^2 &= c^2\sum_{k,l}\frac{[Y,W]|u_k\rangle\,\langle w_k|[Y,W]|u_l\rangle\,\langle w_l|}{\lambda_k\lambda_l}.\end{aligned} \tag{8.7}$$

N is large through the terms of small λ_k, one per low mode: a large part of low rank, which GCR removes in about as many extra steps as there are such modes. N^2 would be large through the terms where both λ_k and λ_l are small, and each of them carries the matrix element $\langle w_k|[Y,W]|u_l\rangle$ between two low modes. By Eq. (8.6) this matrix element is small: a low mode carries P_- at the wall $s=0$ and P_+ at $s=L_s-1$, so $[Y,W]|u_l\rangle$ carries the opposite chirality on each wall, and its overlap with another low mode comes only from the bulk of the fifth dimension, where the low modes have decayed. The left eigenvectors are localized in the same way, since $\Gamma_5 D\Gamma_5 = D^\dagger$ up to $O(c)$. So N^2 is small to the extent that

$$c\,\big|\langle w_k|[Y,W]|u_l\rangle\big| \ll |\lambda_k| \qquad \text{for the low modes } k,l, \tag{8.8}$$

and then $(1-N)^{-1} = 1 + N + N^2 + \cdots \approx 1+N$: GCR needs only a few more steps than for the identity. This is an argument, not a bound: the matrix elements have not been computed, and the iteration counts are the test. On 16^4 it needs 2–3 iterations more than the solve with D; on 12^4 in setup S (stout links with $\rho = 0.1$ and three steps, $M_5 = 1$; note Table 3), GCR(2) stalls on the residual right-hand side of HMC (note, Sec. V.B.2).

8.2 On the even sites (note Sec. IV.B)

The solves of HMC are on the even sites. Since Γ_5 acts within a site it commutes with π and ι, and with Eq. (2.5)

$$S^{-\dagger} = \pi D^{-\dagger}\iota \approx \pi\Gamma_5 D^{-1}\Gamma_5\iota = \Gamma_5\,\pi D^{-1}\iota\,\Gamma_5 = \Gamma_5 S^{-1}\Gamma_5. \tag{8.9}$$

The solve of $D^\dagger|w\rangle = |r\rangle$ is Algorithm 8.

Algorithm 8: Solve with $D^\dagger$ through the multigrid of D

input: source vector $|r\rangle = (|r_e\rangle, |r_o\rangle)$
output: $|w\rangle = D^{-\dagger}|r\rangle$
require: cycle $\mathcal{M}$ of $\tilde{A}_R$ (Algorithm 5), the same setup as for D

1 $|r'_e\rangle := |r_e\rangle - D^\dagger_{oe}(C^\dagger_o)^{-1}|r_o\rangle$ `/* then` $S^\dagger|w_e\rangle = |r'_e\rangle$ `*/`
2 solve $\mathcal{K}^\dagger|z\rangle = |r'_e\rangle$ by GCR (Algorithm 7); its preconditioner maps $|t\rangle$ to
3 $\quad |t_1\rangle := (S_{\rm PV}C^{-1}_{{\rm PV},e})^\dagger\,\Gamma_5|t\rangle$ `/* right-hand side of` $\tilde{A}_R$ `*/`
4 $\quad |t_2\rangle := C_e^{-1}\mathcal{M}|t_1\rangle$ `/* one cycle:` $\approx S^{-1}\Gamma_5|t\rangle$ `*/`
5 $\quad C_e^\dagger\,\Gamma_5|t_2\rangle$ `/*` $\approx C_e^\dagger\Gamma_5 S^{-1}\Gamma_5|t\rangle$ `*/`
6 $|w_e\rangle := (C_e^\dagger)^{-1}|z\rangle$
7 $|w_o\rangle := (C_o^\dagger)^{-1}\big(|r_o\rangle - D^\dagger_{eo}|w_e\rangle\big)$

Line 2 follows from line 6: with $|w_e\rangle = (C_e^\dagger)^{-1}|z\rangle$, $S^\dagger|w_e\rangle = S^\dagger(C_e^\dagger)^{-1}|z\rangle = \mathcal{K}^\dagger|z\rangle$, since $\mathcal{K} = C_e^{-1}S$. The approximation for S^{-1} in lines 3–4 comes from Eq. (3.1): $S^{-1} = C_e^{-1}\tilde{A}_R^{-1}(S_{\rm PV}C^{-1}_{{\rm PV},e})^\dagger$. In line 3, $(S_{\rm PV}C^{-1}_{{\rm PV},e})^\dagger\Gamma_5|t\rangle$ is the right-hand side of $\tilde{A}_R$ for the source $\iota\Gamma_5|t\rangle$ (no odd part), so it is applied directly as one adjoint Schur factor of $D_{\rm PV}$. Each GCR step costs one cycle, as for D.

8.3 Why D_- cannot be inverted (note Sec. IV.C)

The exact relation of Sec. 1.5 contains D_-, and one could try to solve with $D^\dagger$ through D exactly by inverting D_-. Rewrite it:

$$D_- = cW - 1 = c\Big(W - \frac{1}{c}\Big), \tag{8.10}$$

c times a Wilson operator of mass $-M_5 - 1/c$. By Eq. (1.5) with $M_5 \to M_5 + 1/c$, its free eigenvalues are

$$c\Big[-M_5 - \frac{1}{c} + \sum_\mu(1-\cos p_\mu) \pm i\Big(\sum_\mu \sin^2 p_\mu\Big)^{1/2}\Big]. \tag{8.11}$$

For $M_5 = 1$, $c = 0.5$ the real part of the bracket runs from -3 to 5; near real part 0 the imaginary parts have both signs. The spectrum surrounds the origin. A Krylov method needs a polynomial p with $p(0) = 1$ that is small on the spectrum, and on a set around the origin no low-degree polynomial does that (by the maximum principle $|p|$ on a curve around 0 is at least $|p(0)| = 1$). GCR(8) on D_- indeed stops at $|r|^2/|b|^2 \approx 0.74$ after 20000 iterations. The preconditioner of Sec. 8.1 never inverts D_-.

9 Why the method converges (note Sec. V.D)

The outer solver works on $\mathcal{A} \equiv \tilde{A}\mathcal{M}$, the operator times the cycle. The note asks which property of $\mathcal{A}$ explains the convergence it measures. The tools below need $\mathcal{A}$ to be one fixed linear map, so in this section the coarse solves are exact; with Krylov coarse solves stopped at 10^{-2} the cycle changes from one application to the next.

9.1 The field of values and the bound for GCR

Definition. The field of values of a matrix $\mathcal{A}$ is the set of Rayleigh quotients,

$$W(\mathcal{A}) = \Big\{\frac{\langle z|\mathcal{A}|z\rangle}{\langle z|z\rangle} \;:\; |z\rangle \neq 0\Big\}. \tag{9.1}$$

It contains the spectrum: for an eigenvector, $\mathcal{A}|x\rangle = \lambda|x\rangle$ gives $\langle x|\mathcal{A}|x\rangle/\langle x|x\rangle = \lambda$. Its leftmost point is set by the Hermitian part $\mathcal{H} = \frac{1}{2}(\mathcal{A} + \mathcal{A}^\dagger)$. Since $\langle z|\mathcal{A}^\dagger|z\rangle = \langle z|\mathcal{A}|z\rangle^*$,

$$\Re\langle z|\mathcal{A}|z\rangle = \tfrac{1}{2}\big(\langle z|\mathcal{A}|z\rangle + \langle z|\mathcal{A}|z\rangle^*\big) = \langle z|\mathcal{H}|z\rangle \;\geq\; \lambda_{\min}(\mathcal{H})\,\langle z|z\rangle, \tag{9.2}$$

so $\min \Re W(\mathcal{A}) = \lambda_{\min}(\mathcal{H})$.

One step of GCR. Take the residual $|r\rangle$ and the new direction $\mathcal{M}|r\rangle$, so $|q\rangle = \tilde{A}\mathcal{M}|r\rangle = \mathcal{A}|r\rangle$. Minimizing over β alone,

$$\min_{\beta} \big\| |r\rangle - \beta|q\rangle \big\|^2 = \langle r|r\rangle - \frac{|\langle q|r\rangle|^2}{\langle q|q\rangle} \le \langle r|r\rangle - \frac{\big(\Re\langle r|\mathcal{A}|r\rangle\big)^2}{\|\mathcal{A}\|^2\langle r|r\rangle}, \tag{9.3}$$

using $|\langle q|r\rangle| \ge |\Re\langle r|\mathcal{A}|r\rangle|$ and $\langle q|q\rangle \le \|\mathcal{A}\|^2\langle r|r\rangle$. If $W(\mathcal{A})$ lies in the right half-plane, $c \equiv \lambda_{\min}(\mathcal{H}) > 0$ and

$$\frac{\|r_{\text{new}}\|^2}{\|r\|^2} \le 1 - \frac{c^2}{\|\mathcal{A}\|^2} < 1. \tag{9.4}$$

GCR(k) does at least as well: its residual is orthogonal to the kept $|q_j\rangle$, so it minimizes over their span together with $\mathcal{A}|r\rangle$, which contains the one-direction choice above. This is the bound of Eisenstat, Elman and Schultz: every step reduces the residual, for every k, including $k = 0$.

When W contains the origin. Then some $|r\rangle$ has $\langle r|\mathcal{A}|r\rangle = 0$. If such an $|r\rangle$ is reached and the kept directions do not help, $\langle q|r\rangle$ can vanish, so $\beta = 0$, the residual does not change, and with a fixed cycle the next step repeats the same direction. The bound gives nothing, and convergence has to come from elsewhere. The note finds that the coarse correction pushes W across the origin, and that the spectrum stays away from it.

9.2 The spectrum and eigenvalue condition numbers

The spectrum lies inside W, so it can stay far from the origin while W does not. For a non-normal $\mathcal{A}$ the eigenvalues alone do not bound the residual: the eigenvectors are not orthogonal. Let $|x\rangle$ and $\langle y|$ be the normalized right and left eigenvectors of λ. The spectral projector onto λ is

$$P_\lambda = \frac{|x\rangle\langle y|}{\langle y|x\rangle}, \qquad \|P_\lambda\| = \frac{\|x\|\,\|y\|}{|\langle y|x\rangle|} = \frac{1}{|\langle y|x\rangle|} \equiv \kappa_\lambda, \tag{9.5}$$

the condition number of λ. It is 1 when $|x\rangle$ and $|y\rangle$ coincide (a normal matrix) and large when they are nearly orthogonal.

9.3 Pseudospectra and the bound for GMRES

Definition. The ε-pseudospectrum is the set of z where $\sigma_{\min}(z - \mathcal{A}) < \varepsilon$, i.e. where $\|(z - \mathcal{A})^{-1}\| > 1/\varepsilon$. For a normal matrix it is the ε-neighborhood of the spectrum; for a non-normal one it can be much larger.

The bound. Full GMRES picks the best polynomial with $p_k(0) = 1$: $|r_k\rangle = p_k(\mathcal{A})|r_0\rangle$ with $\|r_k\|$ minimal, so for any such polynomial $\|r_k\|/\|r_0\| \le \|p_k(\mathcal{A})\|$. Let a contour Γ of length L enclose the spectrum but not the origin, with $\sigma_{\min}(z - \mathcal{A}) \ge \varepsilon$ on Γ. By Cauchy's formula for a matrix,

$$\begin{aligned} p_k(\mathcal{A}) &= \frac{1}{2\pi i} \oint_\Gamma p_k(z)\,(z - \mathcal{A})^{-1}\,dz, \\ \|p_k(\mathcal{A})\| &\le \frac{1}{2\pi}\, L \max_{z\in\Gamma} |p_k(z)| \max_{z\in\Gamma} \|(z - \mathcal{A})^{-1}\| \le \frac{L}{2\pi\varepsilon} \max_{z\in\Gamma} |p_k(z)|. \end{aligned} \tag{9.6}$$

A circle. Take Γ the circle $|z-c|=r$ with $r<c$ (it excludes the origin) and $p_k(z)=(1-z/c)^k$. On the circle $|1-z/c|=|c-z|/c=r/c$ and $L=2\pi r$, so

$$\frac{\|r_k\|}{\|r_0\|} \le \frac{r}{\varepsilon}\Big(\frac{r}{c}\Big)^k : \tag{9.7}$$

convergence at the rate r/c per iteration, with the constant r/ε set by how far the pseudospectrum reaches.

An outlier. An eigenvalue λ_o outside the circle is removed by a factor of the polynomial, $p_k(z)=(1-z/\lambda_o)(1-z/c)^{k-1}$, which vanishes at λ_o. For a simple λ_o, $p_k(\mathcal{A})P_o = p_k(\lambda_o)P_o = 0$, so $p_k(\mathcal{A}) = p_k(\mathcal{A})(I-P_o)$, and the integral needs only the part of the spectrum inside the circle:

$$\begin{aligned} \|(z-\mathcal{A})^{-1}(I-P_o)\| &\le \|(z-\mathcal{A})^{-1}\|\,\|I-P_o\| \le \frac{\kappa_o}{\varepsilon}, \\ |1-z/\lambda_o| = \frac{|\lambda_o - z|}{|\lambda_o|} &\le \frac{|\lambda_o - c| + r}{|\lambda_o|} \qquad (|z-c|=r), \\ \frac{\|r_k\|}{\|r_0\|} &\le \kappa_o \frac{r}{\varepsilon}\frac{|\lambda_o-c|+r}{|\lambda_o|}\Big(\frac{r}{c}\Big)^{k-1}, \end{aligned} \tag{9.8}$$

using $\|I-P_o\| = \|P_o\| = \kappa_o$ for an oblique projector. The outlier costs one iteration and a constant factor, not the rate.

Checking a contour with finitely many points. $\sigma_{\min}$ is 1-Lipschitz: $|\sigma_{\min}(z-\mathcal{A}) - \sigma_{\min}(z'-\mathcal{A})| \le \|(z-z')I\| = |z-z'|$. If it is at least ε_0 at points spaced by h on Γ, it is at least $\varepsilon_0 - h/2$ everywhere on Γ. This is why the note can state the bound on its dense circles and only an estimate on the contours with 16 points.

9.4 Truncated GCR with inexact coarse solves

The bounds above are for a fixed $\mathcal{M}$ and, for the pseudospectra, for full GMRES. The solver runs GCR(2) with coarse solves to 10^{-2}, so $\mathcal{M}$ changes from step to step and neither bound applies directly. The note checks this case numerically (Sec. V.D.4). The one mechanism of failure that a fixed cycle allows is the repeated direction of the previous subsection, a plateau of the residual.

10 Choices of the detail

Each step above has alternatives. The list gives them with the default of the code first, the parameter that selects them, and the section of the note that compares them where it does.

- The outer system (`operator_type`)
  - $\tilde{A}_R$ (`PV_right`): default; does not see Neff's α, the correction is exact on its span
  - $\tilde{A}_L^{(\alpha)}$ (`PV_left`): sees α; the runs of the note use it at $\alpha = 0.6$
  - CGNE on $\mathcal{K}^\dagger\mathcal{K}$ (`Normal_left`): the comparison without the multigrid (note Sec. V.B.1)
- The outer solver
  - flexible GCR(k) with $k=2$ (`number_of_kept_directions`): default
  - larger k: at most one iteration fewer in the median on 12^4 (note Sec. V.D.4)
- The prolongation basis
  - both lifts $\{|s_j\rangle, |\hat{t}_j\rangle\}$, split by Γ_5-chirality: fixed in the code

  - $\{|s_j\rangle, D|s_j\rangle\}$ or $\{|s_j\rangle\}$ alone: compared in the note (Sec. V.C.2)

- The Pauli–Villars coarse solve (`Coarse_Solver_PV`)
  - CG on the regularized normal equation, $\tau = 10^{-3}$ (`solver_type: CG`, `tikhonov_epsilon`): default
  - BiCGStab or GCR on the exact adjoint, without the shift (`solver_type: BiCGStab`, `GCR`): the note compares it with the default (Sec. V.C.3)
  - MINRES on the Σ-Hermitian form, exact only for $c = 0$ (`solver_type: MINRES`): kept in the code, not compared in the note

- The quark coarse solve (`Coarse_Solver_D`)
  - BiCGStab on the even-odd coarse Schur complement, to $|r|^2/|b|^2 < 10^{-2}$: default
  - with levels below (`super_block_size`): on 24^4 at $m = 0.001$ about 50 instead of 2000 applications of $\mathcal{D}_c$ per correction (Sec. 4)

- The smoother (`Smoother`)
  - Chebyshev, $\nu = 8$ steps on $[\lambda_{hi}/10, \lambda_{hi}]$: default, for setup S
  - $\nu = 12$ on $[\lambda_{hi}/30, \lambda_{hi}]$: setup B (note Table 3; the sensitivity in Sec. V.C.4)

- The setup (`Setup`)
  - filter: 16 vectors, 2 rounds; bootstrap: 2 rounds of 4 FGMRES steps: default
  - 32 test vectors (16 lifts of each kind, `number_of_test_vectors`); 4^4 blocks in the runs (`block_size`, no default)

- The solve with $D^\dagger$
  - GCR on $\mathcal{K}^\dagger$ preconditioned by $\Gamma_5 D^{-1} \Gamma_5$, the setup of D: default (Algorithm 8)
  - a second setup for $D^\dagger$: avoided, since it would double the cost of the setup (Sec. 8)

11 Map to the code

step	where
C, C^{-1}, hops, Γ_5, Q on the device	`src/backend/cuda_bridge/mdwf.rs`; `cuda_bridge/include/fermion/`
even-odd forms, $\tilde{A}_R$, $\tilde{A}_L$ (`PvRatio`, `SchurFactor`, `AfterQ`)	`src/fermion/even_odd.rs`
GCR, FGMRES, BiCGStab, MINRES, CG	`src/solver/`
Chebyshev smoother, two-grid cycle	`src/solver/preconditioner/multigrid/cycle.rs`; `src/backend/cuda_bridge/multigrid.rs`
factorized correction	`src/solver/preconditioner/multigrid/correction.rs`; `src/backend/cuda_bridge/correction.rs`
$\mathcal{P}$: fold, restrict, prolong, CholeskyQR2	`src/backend/cuda_bridge/coarse.rs`; `cuda_bridge/include/multigrid/fold.cuh`
coarse operators by probing	`CudaMdwf::coarse_blocks` in `src/backend/cuda_bridge/mdwf.rs`
levels below	`src/backend/cuda_bridge/levels.rs`
filter, bootstrap	`low_modes` in `src/solver/preconditioner/multigrid/mdwf_even_odd.rs`; `bootstrap` in `src/backend/cuda_bridge/multigrid.rs`
solve with $D^\dagger$	`adjoint_by_gamma5` in `src/fermion/even_odd.rs`